Ilja Safronovs

R&D Engineer — 3D reconstruction & computer vision · Giraffe360 · Riga, Latvia

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I work on 3D reconstruction and computer vision: Gaussian splatting pipelines, benchmarking systems that measure every change, and continuous research that turns the best new papers into production code. Financial Engineering student at RTU.

EXPERIENCE

Giraffe360 — R&D Engineer — Riga · 2025 – present

Solved blurry windows, outdoor views and mirrors in 3DGS video. instagram.com/p/DZsNRB8o772
This was a long-standing quality problem. I found the root cause (wrong depth behind windows and walls), handled depth differently inside and outside the room, and added a training-time correction. 700+ controlled experiments back it. It runs in production and was publicly showcased by Giraffe360.

Shipped the looped drone video, from idea to production. instagram.com/giraffe360/reel/Dd9ODBVIHhE
I proposed the feature, agreed its look with the designers, developed the camera-path algorithm and shipped it to production. A customer-requested feature, launched by Giraffe360 in Oct 2026.

Research to production: influence accumulation, plus Analytic-Splatting in gsplat.
Influence accumulation measures each Gaussian's contribution to the image and relocates the least useful ones during training; it is in production. Analytic-Splatting (exact per-pixel Gaussian integration) is implemented in gsplat at CUDA level. It reduces aliasing and is used in a downsampling algorithm.

Built tools to measure 3DGS quality against known ground truth.
Validation tools and ground-truth datasets, to measure quality against known truth; an experiment runner for agent-driven research loops on local GPUs; ClearML GPU runs.

Led a small team building outdoor splats and orbital videos.
Led a small team that built exterior (outdoor) splats and orbital outdoor videos, now in production.

Mentored a junior engineer on a new depth algorithm.
The junior engineer built a new depth algorithm with me (mostly his work): minima matching plus OMNI-DC to densify LiDAR depth. Now the best method; the team uses it.

Led the pipeline restructuring and created a 3DGS monorepo.
Led the pipeline restructuring and created a monorepo for all 3DGS extensions and client apps: ~3000-line files split, duplication removed, defects fixed. Enabled the half-floor camera fix.

Built an embeddable Rust/wgpu splat viewer and a comparison tool.
Embeddable Rust/wgpu splat viewer (host control, PLY comparison); multi-view comparison tool.

Prototyped mirror planes and a fly-in dollhouse video intro.
Working prototypes: mirror planes with correct reflections; a fly-in "dollhouse" video intro.

Worked on mesh generation and point-count reduction for viewers.
Mesh generation from splats, combined with Gaussian splats; point-count reduction for viewers.

Made diffusion-based artifact fixing runnable on shared GPUs.
Diffusion-based artifact fixing (Difix, NVIDIA Fixer) made runnable on shared GPUs.

Three hackathon ideas recognized; the looped drone shipped.
3 ideas recognized at an internal hackathon (top 20, 2 mentions); the looped drone became a shipped feature.

Built an MCP server for the team task board.
An MCP server so AI agents can read and write the team task board (Node.js).

Earlier: a small LLM service that wrote tour narration.
Jul–Oct 2025: a small LLM service that wrote tour narration (OpenAI API, AWS, S3).
Giraffe360 makes cameras and software for real-estate virtual tours.

EDUCATION

Riga Technical University — Professional Bachelor, Financial Engineering 2024 – 2028 (expected)

Maths, statistics and programming applied to finance.

Qualification: senior data analysis specialist. Maths, statistics and programming applied to finance.

Coursework: mathematical analysis and algebra, probability and statistics, econometrics, time-series analysis, stochastic analysis, Monte Carlo methods, portfolio analysis, actuarial mathematics, data structures and algorithms, databases/SQL, computer networks, programming (Java, C++).

AI Talent Program (Ponder) Nov 2024 – Jan 2025

Mākslīgā intelekta talantu programma

Selective program in C++, Python, maths and AI.

Selective program: C++, Python, maths and AI development with industry mentors. Led to the Giraffe360 role.

Riga State Classical Gymnasium — Programming and Mathematics track

Rīgas Valsts klasiskā ģimnāzija

State mathematics olympiads and the Small Mathematics University.

Specialised track in programming and mathematics (today named "Programming + AI"). Latvian state mathematics olympiads; completed the Small Mathematics University (Mazā matemātikas universitāte) at the University of Latvia.

PROJECTS

load-balancing-experiments — seven experiments on DNS, queueing and load balancing.

flow — offline workout tracker in Dart/Flutter, in daily use.

monte-carlo-finance — option pricing, volatility and risk models in Python.

Portfolio journey — this tap-driven WebGL site in three.js.

Offline Graphing Calculator — 4th place of 27 in a school competition.

crypto-quant-lab — 59 backtested trading-rule versions on 6 coins.

projects_with_NEAT — neuroevolution agents that learn Flappy Bird and Pong.

RTU_program_parser — finds extra RTU lectures that fit your schedule.

Snake + maze generator — maze generation, terrain and path-finding in Pygame.

bezier-curve-js — interactive Bezier-curve drawing tool.

sort_visualization — 16 sorting algorithms animated in C++/SFML.

SKILLS

3D & vision	Gaussian splatting, 3D reconstruction, computer vision, depth estimation, anti-aliasing, mesh generation, diffusion-based artifact fixing, PyTorch, CUDA kernels
Research	Paper-to-production, benchmarking and evaluation
Languages	Python, C++, Dart/Flutter, JavaScript, Java, SQL
Tools	Linux, Docker, Git/GitLab CI, ClearML, AWS S3, Pygame, Selenium, pandas
Other	OpenAI API / LLM prompting, neuroevolution (NEAT), three.js/WebGL
Spoken	Latvian, English, Russian